

Experiment 1 – Control Structure Statements

Q1. Develop a Java program to calculate the electricity bill based on the number of units consumed using appropriate control statements. Assume the bill is calculated as $\text{Bill} = \text{Units} \times \text{Rate}$, where the rate depends on the slab selected using decision-making statements.

Answer:

```
import java.util.Scanner;

public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int units;

        double rate, bill;

        System.out.print("Enter units consumed: ");

        units = sc.nextInt();

        if (units <= 100)

            rate = 3.0;

        else if (units <= 200)

            rate = 5.0;

        else

            rate = 7.0;

        bill = units * rate;

        System.out.println("Rate per Unit = " + rate);
```

```
System.out.println("Electricity Bill = " + bill);

sc.close();

}

}
```

Q2. Write a Java program to determine whether a given year is a leap year using decision-making statements. A year is considered a leap year if (Year % 400 = 0) OR (Year % 4 = 0 AND Year % 100 ≠ 0).

Answer:

```
import java.util.Scanner;
```

```
public class Main {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        int year;
```

```
        System.out.print("Enter Year: ");
```

```
        year = sc.nextInt();
```

```
        if ((year % 400 == 0) || (year % 4 == 0 && year % 100 != 0))
```

```
            System.out.println(year + " is a Leap Year");
```

```
        else
```

```
            System.out.println(year + " is Not a Leap Year");
```

```
        sc.close();
    }
}
```

Q3. Develop a Java program to display the multiplication table of a given number using loops. Calculate each value using the formula $\text{Table Value} = \text{Number} \times i$, where $i = 1$ to 10 .

Answer:

```
import java.util.Scanner;
```

```
public class Main {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        int num;
```

```
        System.out.print("Enter Number: ");
```

```
        num = sc.nextInt();
```

```
        for (int i = 1; i <= 10; i++) {
```

```
            System.out.println(num + " x " + i + " = " + (num * i));
```

```
        }
```

```
        sc.close();
```

```
    }
```

```
}
```

Q4. Write a Java program to determine the largest among three numbers by comparing the values using nested decision-making statements.

Answer:

```
import java.util.Scanner;
```

```
public class Main {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        int a, b, c;
```

```
        System.out.print("Enter First Number: ");
```

```
        a = sc.nextInt();
```

```
        System.out.print("Enter Second Number: ");
```

```
        b = sc.nextInt();
```

```
        System.out.print("Enter Third Number: ");
```

```
        c = sc.nextInt();
```

```
        if (a > b) {
```

```
            if (a > c)
```

```
                System.out.println("Largest Number = " + a);
```

```
            else
```

```
                System.out.println("Largest Number = " + c);
```

```
        } else {
```

```

        if (b > c)

            System.out.println("Largest Number = " + b);

        else

            System.out.println("Largest Number = " + c);

    }

    sc.close();

}

}

```

Experiment 2 – Functions

Q5. Develop a Java program to calculate the area of a circle using a user-defined function. Use the formula $\text{Area} = \pi \times r^2$.

Answer:

```

import java.util.Scanner;

public class Main {

    static double area(double r) {

        return 3.14 * r * r;

    }

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        double radius;
    }
}

```

```

        System.out.print("Enter Radius: ");

        radius = sc.nextDouble();

        System.out.println("Area of Circle = " + area(radius));

        sc.close();
    }
}

```

Q6. Write a Java program to calculate the factorial of a number using a user-defined function. Use the formula $n! = n \times (n-1) \times (n-2) \times \dots \times 1$.

Answer:

```

import java.util.Scanner;

public class Main {

    static long factorial(int n) {

        long fact = 1;

        for (int i = 1; i <= n; i++) {

            fact = fact * i;

        }

        return fact;

    }

    public static void main(String[] args) {

```

```
Scanner sc = new Scanner(System.in);

int num;

System.out.print("Enter Number: ");

num = sc.nextInt();

System.out.println("Factorial = " + factorial(num));

sc.close();
}
}
```

Q7. Develop a Java program to check whether a given number is prime using a user-defined function. A number is prime if it has exactly two factors (1 and itself).

Answer:

```
import java.util.Scanner;

public class Main {

    static boolean isPrime(int n) {

        if (n <= 1)

            return false;

        for (int i = 2; i <= n / 2; i++) {
```

```

        if (n % i == 0)
            return false;
    }

    return true;
}

public static void main(String[] args) {

    Scanner sc = new Scanner(System.in);

    int num;

    System.out.print("Enter Number: ");
    num = sc.nextInt();

    if (isPrime(num))
        System.out.println(num + " is Prime");
    else
        System.out.println(num + " is Not Prime");

    sc.close();
}
}

```

Q8. Write a Java program to determine the maximum of two numbers using a user-defined function by comparing both numbers.

Answer:

```
import java.util.Scanner;
```

```
public class Main {
```

```
    static int maximum(int a, int b) {
```

```
        if (a > b)
```

```
            return a;
```

```
        else
```

```
            return b;
```

```
    }
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        int num1, num2;
```

```
        System.out.print("Enter First Number: ");
```

```
        num1 = sc.nextInt();
```

```
        System.out.print("Enter Second Number: ");
```

```
        num2 = sc.nextInt();
```

```
        System.out.println("Maximum Number = " + maximum(num1, num2));
```

```
        sc.close();
    }
}
```

Experiment 3 – Constructors

Q9. Develop a Java program to store and display employee details such as Employee ID, Name, and Salary using a default constructor.

Answer:

```
public class Main {

    int empId;

    String name;

    double salary;

    // Default Constructor
    Main() {

        empId = 101;

        name = "Amit";

        salary = 35000;

    }

    void display() {

        System.out.println("Employee ID : " + empId);

        System.out.println("Employee Name : " + name);

        System.out.println("Salary : " + salary);

    }

}
```

```
public static void main(String[] args) {  
    Main emp = new Main();  
    emp.display();  
}  
}
```

Q10. Write a Java program to initialize student details such as Roll Number, Name, and Marks using a parameterized constructor.

Answer:import java.util.Scanner;

```
public class Main {  
  
    int rollNo;  
    String name;  
    double marks;  
  
    // Parameterized Constructor  
    Main(int rollNo, String name, double marks) {  
        this.rollNo = rollNo;  
        this.name = name;  
        this.marks = marks;  
    }  
  
    void display() {  
        System.out.println("Roll Number : " + rollNo);  
        System.out.println("Name : " + name);  
        System.out.println("Marks : " + marks);  
    }  
}
```

```
}

public static void main(String[] args) {

    Scanner sc = new Scanner(System.in);

    int roll = sc.nextInt();

    sc.nextLine();

    String name = sc.nextLine();

    double marks = sc.nextDouble();

    Main s = new Main(roll, name, marks);

    s.display();

    sc.close();

}

}
```

Q11. Develop a Java program to create a copy of an existing Book object by copying all its data members into another object using a copy constructor (or equivalent implementation).

Answer:

```
public class Main {
```

```
    int bookId;
```

```
    String title;
```

```
    double price;
```

```
// Parameterized Constructor
```

```
Main(int id, String t, double p) {
```

```
    bookId = id;
```

```
    title = t;
```

```
    price = p;
```

```
}
```

```
// Copy Constructor
```

```
Main(Main b) {
```

```
    bookId = b.bookId;
```

```
    title = b.title;
```

```
    price = b.price;
```

```
}
```

```
void display() {
```

```
    System.out.println(bookId + " " + title + " " + price);
```

```
}
```

```
public static void main(String[] args) {
```

```
    Main b1 = new Main(1, "Java Programming", 550);
```

```
    Main b2 = new Main(b1);
```

```
    System.out.println("Original Book:");
```

```
    b1.display();
```

```
        System.out.println("Copied Book:");  
        b2.display();  
    }  
}
```

Q12. Design a Java application to maintain product details using both default and parameterized constructors.

Answer:

```
public class Main {
```

```
    int productId;
```

```
    String productName;
```

```
    double price;
```

```
// Default Constructor
```

```
Main() {
```

```
    productId = 101;
```

```
    productName = "Mouse";
```

```
    price = 450;
```

```
}
```

```
// Parameterized Constructor
```

```
Main(int id, String name, double p) {
```

```
    productId = id;
```

```
    productName = name;
```

```
    price = p;
```

```

    }

    void display() {
        System.out.println("Product ID : " + productId);
        System.out.println("Product Name : " + productName);
        System.out.println("Price : " + price);
    }

    public static void main(String[] args) {

        Main p1 = new Main();

        Main p2 = new Main(102, "Keyboard", 900);

        System.out.println("Default Constructor:");
        p1.display();

        System.out.println();

        System.out.println("Parameterized Constructor:");
        p2.display();
    }
}

```

Experiment 4 – Method Overloading

Q13. Develop a calculator application using method overloading to calculate the sum = a + b, sum =

a + b + c, and sum = a + b for decimal values.

Answer:

```
public class Main {
```

```
    // Sum of two integers
```

```
    int sum(int a, int b) {
```

```
        return a + b;
```

```
    }
```

```
    // Sum of three integers
```

```
    int sum(int a, int b, int c) {
```

```
        return a + b + c;
```

```
    }
```

```
    // Sum of two decimal numbers
```

```
    double sum(double a, double b) {
```

```
        return a + b;
```

```
    }
```

```
    public static void main(String[] args) {
```

```
        Main obj = new Main();
```

```
        System.out.println("Sum of 10 and 20 = " + obj.sum(10, 20));
```

```
        System.out.println("Sum of 10, 20 and 30 = " + obj.sum(10, 20, 30));
```

```
        System.out.println("Sum of 10.5 and 20.5 = " + obj.sum(10.5, 20.5));
```

```
    }
```



```
}
```

Q14. Write a Java program to calculate the area of different shapes using method overloading. Use the formulas Square = side², Rectangle = length × breadth, and Circle = $\pi \times r^2$.

Answer:

```
public class Main {
```

```
    // Area of Square
```

```
    int area(int side) {
```

```
        return side * side;
```

```
    }
```

```
    // Area of Rectangle
```

```
    int area(int length, int breadth) {
```

```
        return length * breadth;
```

```
    }
```

```
    // Area of Circle
```

```
    double area(double radius) {
```

```
        return 3.14 * radius * radius;
```

```
    }
```

```
    public static void main(String[] args) {
```

```
        Main obj = new Main();
```

```
        System.out.println("Area of Square = " + obj.area(5));
```

```

        System.out.println("Area of Rectangle = " + obj.area(6, 4));

        System.out.println("Area of Circle = " + obj.area(5.0));
    }
}

```

Q15. Develop a billing application using method overloading to calculate the total amount using the formula $\text{Total} = \text{Quantity} \times \text{Price}$ and the discounted amount using $\text{Final Amount} = \text{Total} - (\text{Total} \times \text{Discount} / 100)$.

Answer:

```

public class Main {

    // Total Amount

    double bill(int quantity, double price) {

        return quantity * price;
    }

    // Discounted Bill

    double bill(int quantity, double price, double discount) {

        double total = quantity * price;

        return total - (total * discount / 100);
    }

    public static void main(String[] args) {

        Main obj = new Main();

        System.out.println("Total Amount = " + obj.bill(5, 100));
    }
}

```

```
        System.out.println("Final Amount after Discount = " + obj.bill(5, 100, 10));
    }
}
```

Q16. Write a Java program to calculate Simple Interest using method overloading. Use the formula $SI = (Principal \times Rate \times Time)/100$.

Answer:

```
public class Main {

    // SI using integer values
    double interest(int p, int r, int t) {
        return (p * r * t) / 100.0;
    }

    // SI using decimal rate
    double interest(double p, double r, double t) {
        return (p * r * t) / 100.0;
    }

    public static void main(String[] args) {

        Main obj = new Main();

        System.out.println("Simple Interest = " + obj.interest(10000, 5, 2));
        System.out.println("Simple Interest = " + obj.interest(15000.0, 7.5, 3.0));
    }
}
```

Experiment 5 – Vector

Q17. Develop a Java program to maintain a list of student names using the Vector class by performing insertion, deletion, searching, and display operations.

Answer:

```
import java.util.Vector;

public class Main {

    public static void main(String[] args) {

        Vector<String> students = new Vector<>();

        // Insert

        students.add("Amit");

        students.add("Rahul");

        students.add("Sneha");

        // Display

        System.out.println("Student List:");

        System.out.println(students);

        // Search

        if (students.contains("Rahul"))

            System.out.println("Rahul Found");

        else

            System.out.println("Rahul Not Found");
```

```

// Delete

students.remove("Rahul");

System.out.println("After Deletion:");

System.out.println(students);

}

}

```

Q18. Write a Java application to maintain book records using the Vector class and perform add, search, remove, and display operations.

Answer:

```
import java.util.Vector;
```

```

public class Main {

    public static void main(String[] args) {

        Vector<String> books = new Vector<>();

        // Add Books

        books.add("Java");

        books.add("Python");

        books.add("C Programming");

        System.out.println("Book List:");

        System.out.println(books);

        // Search
    }
}

```

```

if (books.contains("Python"))

    System.out.println("Python Book Found");

else

    System.out.println("Python Book Not Found");


// Remove

books.remove("Python");


System.out.println("After Removing Python:");

System.out.println(books);

}

}

```

Q19. Develop a Java program to store employee IDs using the Vector class and display all IDs.

Answer:

```
import java.util.Vector;
```

```
public class Main {
```

```
    public static void main(String[] args) {
```

```
        Vector<Integer> empIds = new Vector<>();
```

```
        empIds.add(101);
```

```
        empIds.add(102);
```

```
        empIds.add(103);
```

```
        empIds.add(104);
```

```

        System.out.println("Employee IDs:");

        for (int id : empIds) {

            System.out.println(id);

        }

    }

}

```

Q20. Create a shopping cart application using the Vector class to add, remove, and display purchased items.

Answer:

```
import java.util.Vector;
```

```

public class Main {

    public static void main(String[] args) {

        Vector<String> cart = new Vector<>();

        // Add Items

        cart.add("Laptop");

        cart.add("Mouse");

        cart.add("Keyboard");

        System.out.println("Shopping Cart:");

        System.out.println(cart);

        // Remove Item
    }

}

```

```
        cart.remove("Mouse");

        System.out.println("After Removing Mouse:");

        System.out.println(cart);
    }
}
```

Experiment 6 – String Operations

Q21. Develop a Java program to count the number of vowels in a given string by checking each character against A, E, I, O, U.

Answer:

```
import java.util.Scanner;

public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a String: ");

        String str = sc.nextLine();

        int count = 0;

        str = str.toLowerCase();

        for (int i = 0; i < str.length(); i++) {

            char ch = str.charAt(i);
```



```

        if (ch == 'a' || ch == 'e' || ch == 'i' ||
            ch == 'o' || ch == 'u') {
            count++;
        }
    }

    System.out.println("Number of Vowels = " + count);

    sc.close();
}
}

```

Q22. Write a Java program to determine whether a given string is a palindrome by reversing the string and comparing it with the original string.

Answer:import java.util.Scanner;

```

public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a String: ");

        String original = sc.nextLine();

        String reverse = "";
    }
}

```

```

    for (int i = original.length() - 1; i >= 0; i--) {
        reverse = reverse + original.charAt(i);
    }

    if (original.equalsIgnoreCase(reverse))
        System.out.println("Palindrome");
    else
        System.out.println("Not Palindrome");

    sc.close();
}
}

```

Q23. Develop a Java program to compare two strings entered by the user using appropriate string comparison methods.

Answer:

```
import java.util.Scanner;
```

```

public class Main {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter First String: ");
        String s1 = sc.nextLine();

        System.out.print("Enter Second String: ");

```

```
String s2 = sc.nextLine();

if (s1.equals(s2))
    System.out.println("Strings are Equal");
else
    System.out.println("Strings are Not Equal");

int result = s1.compareTo(s2);

if (result == 0)
    System.out.println("compareTo() Result: Equal");
else if (result > 0)
    System.out.println("First String is Greater");
else
    System.out.println("Second String is Greater");

sc.close();
}
}
```

Q24. Write a Java program to perform string operations such as finding length, converting to uppercase, converting to lowercase, and extracting a substring.

Answer:import java.util.Scanner;

```
public class Main {

    public static void main(String[] args) {
```

```

Scanner sc = new Scanner(System.in);

System.out.print("Enter a String: ");

String str = sc.nextLine();

System.out.println("Length = " + str.length());

System.out.println("Uppercase = " + str.toUpperCase());

System.out.println("Lowercase = " + str.toLowerCase());

if (str.length() >= 5)

    System.out.println("Substring = " + str.substring(0, 5));

else

    System.out.println("Substring = " + str);

sc.close();
}
}

```

Experiment 7 – Multilevel Inheritance

Q25. Develop a Java application using multilevel inheritance to display details of a Person, Employee, and Manager by inheriting common properties through a three-level class hierarchy.

Answer:import java.util.Scanner;

```

class Person {

    String name;

    int age;

```

```
void getPersonData(Scanner sc) {  
    System.out.print("Enter Name: ");  
    name = sc.nextLine();  
  
    System.out.print("Enter Age: ");  
    age = sc.nextInt();  
}  
}
```

```
class Employee extends Person {  
    int empId;  
    double salary;  
  
    void getEmployeeData(Scanner sc) {  
        System.out.print("Enter Employee ID: ");  
        empId = sc.nextInt();  
  
        System.out.print("Enter Salary: ");  
        salary = sc.nextDouble();  
    }  
}
```

```
class Manager extends Employee {  
    String department;
```

```

void getManagerData(Scanner sc) {
    sc.nextLine();
    System.out.print("Enter Department: ");
    department = sc.nextLine();
}

void display() {
    System.out.println("\n----- Manager Details -----");
    System.out.println("Name      : " + name);
    System.out.println("Age       : " + age);
    System.out.println("Employee ID: " + empId);
    System.out.println("Salary    : " + salary);
    System.out.println("Department : " + department);
}
}

public class Main {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        Manager m = new Manager();

        m.getPersonData(sc);
        m.getEmployeeData(sc);
        m.getManagerData(sc);
    }
}

```

```
m.display();

sc.close();
}
}
```

Q26. Write a Java program using multilevel inheritance to display information of Student, Undergraduate Student, and Postgraduate Student.

Answer:

```
import java.util.Scanner;
```

```
class Student {

    String name;

    int rollNo;

    void getStudentData(Scanner sc) {

        System.out.print("Enter Name: ");

        name = sc.nextLine();

        System.out.print("Enter Roll Number: ");

        rollNo = sc.nextInt();

    }

}
```

```
class UndergraduateStudent extends Student {

    String ugCourse;
```

```

void getUGData(Scanner sc) {
    sc.nextLine();
    System.out.print("Enter UG Course: ");
    ugCourse = sc.nextLine();
}
}

class PostgraduateStudent extends UndergraduateStudent {
    String pgCourse;

    void getPGData(Scanner sc) {
        System.out.print("Enter PG Course: ");
        pgCourse = sc.nextLine();
    }

    void display() {
        System.out.println("\n----- Student Details -----");
        System.out.println("Name      : " + name);
        System.out.println("Roll Number : " + rollNo);
        System.out.println("UG Course  : " + ugCourse);
        System.out.println("PG Course  : " + pgCourse);
    }
}

public class Main {

```



```

public static void main(String[] args) {

    Scanner sc = new Scanner(System.in);

    PostgraduateStudent s = new PostgraduateStudent();

    s.getStudentData(sc);
    s.getUGData(sc);
    s.getPGData(sc);

    s.display();

    sc.close();
}
}

```

Q27. Develop a banking application using multilevel inheritance to represent Account → Savings Account → Premium Savings Account.

Answer:

```

import java.util.Scanner;

class Account {
    int accountNo;
    String holderName;

    void getAccountData(Scanner sc) {
        System.out.print("Enter Account Number: ");
        accountNo = sc.nextInt();

        sc.nextLine();

        System.out.print("Enter Account Holder Name: ");
    }
}

```

```

        holderName = sc.nextLine();
    }
}

class SavingsAccount extends Account {
    double balance;

    void getBalance(Scanner sc) {
        System.out.print("Enter Balance: ");
        balance = sc.nextDouble();
    }
}

class PremiumSavingsAccount extends SavingsAccount {
    double bonus = 500;

    void display() {
        System.out.println("\n----- Account Details -----");
        System.out.println("Account Number : " + accountNo);
        System.out.println("Holder Name   : " + holderName);
        System.out.println("Balance      : " + balance);
        System.out.println("Bonus        : " + bonus);
        System.out.println("Total Balance : " + (balance + bonus));
    }
}

public class Main {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        PremiumSavingsAccount acc = new PremiumSavingsAccount();

        acc.getAccountData(sc);
        acc.getBalance(sc);

        acc.display();

        sc.close();
    }
}

```